

Software Design Document (Checkpoint-2)

Smart Classroom Monitoring System for Schedule Compliance & Energy Awareness

1. Document Control

Field	Details
Project Title	Smart Classroom Monitoring System for Schedule Compliance & Energy Awareness
Group ID	FYP-2026-G3
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Part B: Updated Architecture (FYP-T5)

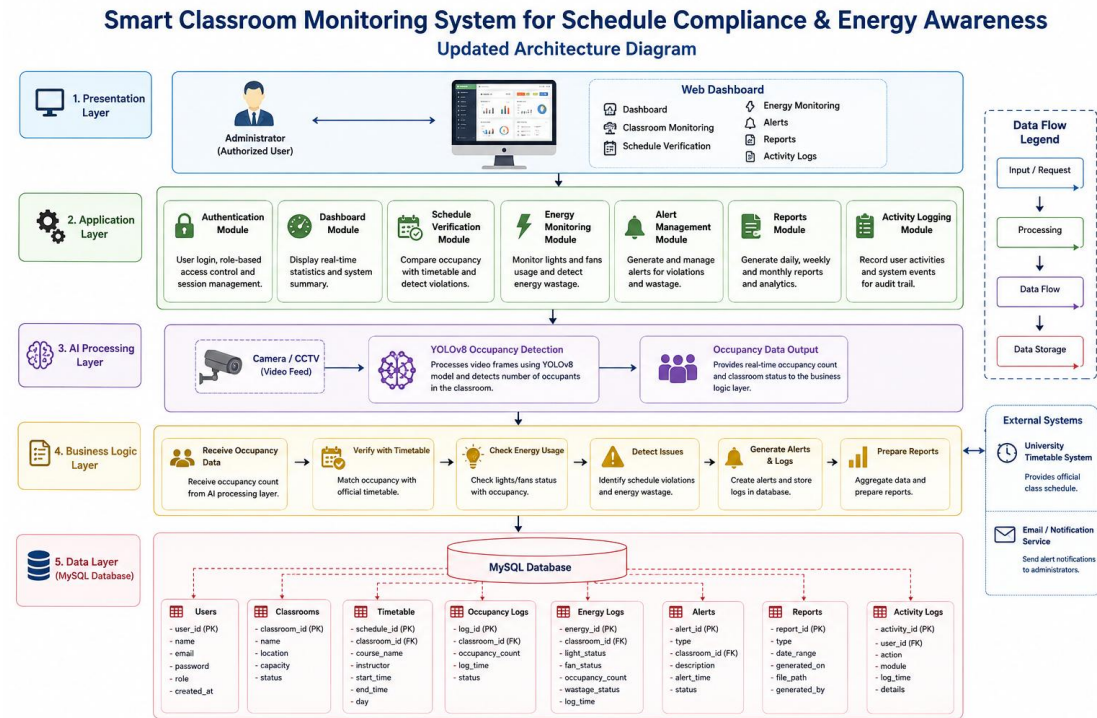
1. Purpose

This document records the architectural status of the Smart Classroom Monitoring System during Checkpoint-2. It identifies refinements made after the completion of the Software Requirements Specification (SRS), Checkpoint-1 architecture, supervisor feedback, and prototype development. The purpose of this update is to ensure that the system architecture accurately reflects the current implementation approach and supports future development activities.

2. Architecture Update Summary

Item	Details
Architecture Status	☒ Minor Refinement
Reason for Update	Minor improvements were made after finalizing the Software Design Document and prototype planning. The core architecture remains unchanged, while additional modules such as Activity Logging, Energy Monitoring, and Alert Management were refined to improve system functionality and maintainability.
Affected Layers / Components	AI Processing Layer, Business Logic Layer, Dashboard Layer, Database Layer
Impact on Technology Stack	No changes were made to the selected technologies. The system continues to use HTML, CSS, Bootstrap, JavaScript, PHP, MySQL, and YOLOv8 for implementation.
Impact on Risks / Security	Security was strengthened by introducing role-based access control, secure session management, password hashing, activity logging, and improved input validation mechanisms.

2. Updated Architecture Diagram



Architecture Description

The updated architecture follows a layered design consisting of five major layers:

Presentation Layer

Provides the web-based user interface including Login, Dashboard, Classroom Monitoring, Reports, Alerts, and Schedule Verification screens.

Application Layer

Processes user requests, manages authentication, controls dashboard operations, and coordinates communication between system modules.

AI Processing Layer

Receives classroom camera frames and processes them using the YOLOv8 object detection model to determine classroom occupancy.

Business Logic Layer

Compares occupancy results with the official timetable, detects schedule violations, monitors classroom energy usage, generates alerts, and prepares reports.

Database Layer

Stores user information, classroom records, schedules, occupancy logs, energy monitoring data, alerts, reports, and activity logs using MySQL.

4. Updated Component Mapping

Component	Change / Refinement	Reason	Impact on Design / Implementation
Authentication Module	Added secure session handling and role-based authorization.	Improve system security.	Prevents unauthorized access.
Camera Capture Module	Optimized frame capture frequency.	Improve processing efficiency.	Reduces unnecessary processing load.
Occupancy Detection Module	Updated to use YOLOv8 detection workflow.	Improve detection accuracy.	Faster and more reliable occupancy detection.
Schedule Verification Module	Added automatic compliance checking.	Improve classroom monitoring.	Generates schedule violation records automatically.
Energy Monitoring Module	Enhanced energy wastage detection logic.	Improve energy awareness.	Detects unnecessary use of classroom appliances.
Alert Management Module	Added centralized alert generation and storage.	Improve administrator response.	Better monitoring and notification system.
Dashboard Module	Added summary cards and graphical statistics.	Improve usability.	Easier visualization of classroom status.
Reporting Module	Added daily and monthly analytical reports.	Improve decision making.	Provides better monitoring insights.
Activity Logging Module	Newly introduced for audit purposes.	Improve traceability.	Records important user and system activities.

5. Updated Risk / Security Notes

Risk / Security Area	Update	Mitigation / Action
Unauthorized Access	Role-based authentication implemented.	Only authorized users can access administrative features.
Invalid User Input	Input validation added on client and server sides.	Prevents invalid or malicious data submission.
Password Security	Passwords stored as hashed values.	Protects user credentials from unauthorized disclosure.
Session Hijacking	Secure session management implemented.	Sessions expire automatically after logout or inactivity.
Camera Failure	Monitoring status and connection errors recorded.	Administrator is notified when camera connection fails.
Database Failure	Error logging and exception handling implemented.	Prevents data loss and supports troubleshooting.
AI Detection Failure	Detection errors logged and monitoring continues with subsequent frames.	Improves system reliability during temporary failures.
Data Integrity	Primary keys, foreign keys, and database constraints enforced.	